

Introduction to C

This worksheet will give you some practice using the C programming language. Make sure to refer to the slides if you are forgetting any of the syntax or the library functions. You can also raise your hand and ask for help if you are stuck!

1. Create a C program and add the following includes at the top:

```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>
#include <unistd.h>
```

2. Write a simple C program that creates two integers and computes its sum. Then print a message back to the screen, in the format “<int1> plus <int2> equals <sum>”.

Challenge: have the program read the two integers from the user instead of hardcoding them. You can use the following resource to help you https://www.w3schools.com/c/c_user_input.php

3. Save your program in the file `intro_c.c`. What is the command to compile your program into an executable named `intro_c`?

4. In the same program. Define a struct type that represents the current time. It should have fields for the hour, minute, and second. Create a variable of this struct to store the current time (just enter it manually).
5. Write a function that takes in this time struct as an argument and prints it to the screen in the format “HH:MM:SS”.
Hint: use the format specifier `%02d` to pad single-digit numbers with a leading zero.
6. Create a while loop in main that updates the struct every second and prints the new time. Remember there cannot be more than 60 seconds ;). (hint: you’ll need to modify the struct and use the `sleep()` function)
7. Will your program run forever? Discuss with your partner and write down your answer, then press `Ctrl+C` while in your terminal to stop it.

8. Modify your while loop so that it stops when a minute has passed.

Challenge: Replace the while loop logic with a for loop instead.